

Fractional–Integral Separation and Continuous Plateaus in Cyclic Hard-Core Covering Arrays

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September 24, 2026

Abstract

We study binary strength-two covering arrays of index λ on cycle topologies, subject to the hard-core constraint that adjacent coordinates cannot both equal 1. We prove a universal lower bound $LP_\lambda(n) \geq 6\lambda$ for all $n \geq 11$, and show that for every fixed integer $\lambda \geq 1$, the fractional optimum converges exponentially to 6λ , whereas the minimum number of distinct rows satisfies $N_\lambda(n) = \Theta_\lambda(\log n)$. Thus the additive integrality gap diverges logarithmically with dimension. We also present exact, independently checkable finite certificates for selected cycle lengths, including an unbroken ten-length interval $n \in [13, 22]$ on which the index-two optimum remains rigidly invariant at 13. Finally, we formulate the dual column-code capacity $L_\lambda(M)$ and establish a bounded potential-deficit framework for exploring the jump threshold.

Keywords: Covering arrays, forbidden edges, CAFE, hard-core model, integrality gap, Markov chains, transfer operator.

MSC (2020): 05B15, 68R05, 90C10, 94B25.

1 Introduction

Combinatorial interaction testing (CIT) is a fundamental methodology in software verification and hardware validation, aiming to detect faulty interactions among n input parameters using minimal test suites [3, 5]. When all interactions are admissible, the optimal test suite size is formalized by the classical theory of *covering arrays*. However, in real-world engineering systems, physical incompatibilities and safety invariants frequently forbid certain parameter configurations from occurring simultaneously [2].

To mathematically model such constrained testing domains, Danziger, Mendelsohn, Moura, and Stevens [4] introduced *Covering Arrays Avoiding Forbidden Edges* (CAFE). Given a graph of forbidden interactions $G = (V, E)$, an admissible test configuration is an assignment $x \in \{0, 1\}^V$ such that no forbidden edge in E has both endpoints simultaneously active (i.e., $x_u x_v = 0$ for all $(u, v) \in E$). In graph-theoretic terms, the set of admissible test rows corresponds precisely to the independent sets of G , forming the canonical hard-core lattice model [1].

In this paper, we focus on the fundamental cyclic topology $G = C_n$, where n binary factors are arranged along a ring and adjacent factors cannot both be active. While heuristic search algorithms (such as simulated annealing and local repair) have been widely deployed on constrained arrays [2, 3], rigorous structural results on the asymptotic behavior of the linear programming (LP) relaxation versus the integer optimum, as well as exact phase-transition boundaries for cyclic CAFE, have remained largely open.

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1.1 Mathematical Model and Notation

Let $C_n = (V, E)$ denote the undirected cycle graph on vertices $V = \{0, 1, \dots, n-1\}$ with edges $(i, i+1 \pmod n)$. A binary test row $x = (x_0, \dots, x_{n-1}) \in \{0, 1\}^n$ is *legal* if

$$x_i + x_{i+1 \pmod n} \leq 1 \quad \text{for all } i \in \{0, \dots, n-1\}. \quad (1)$$

The set of all legal test configurations is denoted by $\mathcal{L}(C_n)$. It is well known that the cardinality $|\mathcal{L}(C_n)|$ is given by the Lucas numbers $L_n = F_{n-1} + F_{n+1} \approx \phi^n$, where $\phi = (1 + \sqrt{5})/2$.

For a fixed coverage strength $t = 2$ and index $\lambda \geq 1$, a test suite $X \subseteq \mathcal{L}(C_n)$ must cover every legal pairwise state at least λ times:

- For *adjacent pairs* (distance $d = 1$ on C_n), the state $(1, 1)$ is forbidden. The 3 feasible states are $(0, 0), (0, 1), (1, 0)$.
- For *non-adjacent pairs* (distance $d \in \{2, \dots, \lfloor n/2 \rfloor\}$), all 4 binary states $(0, 0), (0, 1), (1, 0), (1, 1)$ are feasible.

The total number of feasible pairwise requirements on C_n is $D_n = 3n + 4 \left(\binom{n}{2} - n \right) = 2n^2 - 3n$.

Definition 1.1 (Integer and Fractional Optima). Let $z_r \in \{0, 1\}$ indicate the selection of row $r \in \mathcal{L}(C_n)$. The integer program (IP) and its linear programming (LP) relaxation are defined by:

$$\begin{aligned} N_\lambda(n) &= \min \sum_{r \in \mathcal{L}(C_n)} z_r \quad \text{s.t.} \quad \sum_{r: (r_i, r_j) = (a, b)} z_r \geq \lambda \quad (\forall \text{ feasible } (i, j, a, b)), \quad z_r \in \{0, 1\}, \\ LP_\lambda(n) &= \min \sum_{r \in \mathcal{L}(C_n)} y_r \quad \text{s.t.} \quad \sum_{r: (r_i, r_j) = (a, b)} y_r \geq \lambda \quad (\forall \text{ feasible } (i, j, a, b)), \quad 0 \leq y_r \leq 1. \end{aligned} \quad (2)$$

The *additive integrality gap* is $G_\lambda(n) = N_\lambda(n) - LP_\lambda(n) \geq 0$. For the baseline index $\lambda = 2$, we abbreviate $LP(n) = LP_2(n)$, $N(n) = N_2(n)$, and $G(n) = G_2(n)$.

1.2 Summary of Main Results

The main contributions of this paper are structured as follows:

1. **Universal Dual Lower Bound (Section 2)**: We establish a local potential inequality on cycle independent sets proving that $LP_\lambda(n) \geq 6\lambda$ for all $n \geq 11$ and all $\lambda \geq 1$. For $\lambda = 2$, any 12-row solution is strictly forced to consist entirely of $\{2, 3\}$ -gap configurations (tight rows).
2. **Asymptotic Fractional Convergence (Section 3)**: By constructing a 2-step gap Markov chain and analyzing a 10-state site transfer operator with Perron–Frobenius spectral gap, we prove that the fractional relaxation exponentially rapidly saturates at the universal theoretical floor:

$$\lim_{n \rightarrow \infty} LP_\lambda(n) = 6\lambda, \quad |LP_\lambda(n) - 6\lambda| = O_\lambda(\theta^n) \quad (0 < \theta < 1).$$

3. **Logarithmic Divergence and Unbounded Gap (Section 4)**: Via a hypercube sphere-packing lower bound and probabilistic construction with pairwise distinct rows, we prove:

$$N_\lambda(n) = \Theta_\lambda(\log n) \implies G_\lambda(n) = \Theta_\lambda(\log n) \longrightarrow \infty.$$

4. **The 10-Integer Invariant Plateau (Section 5)**: On finite cycles, we provide exact, solver-free and mechanical certificates for $n \in \{13, \dots, 22\}$, showing that the index-2 optimum remains rigidly invariant across ten consecutive integers:

$$N(13) = N(14) = \dots = N(22) = 13.$$

5. **Dual Capacity and Phase Transition (Section 6):** We formulate the dual column-code capacity $L_\lambda(M)$, show that C_{22} admits a tight 13-row solution while C_{23} admits no tight 13-row solution, and introduce a potential-deficit budget of at most 23 to govern the non-tight search on C_{23} .

2 Universal Local Potential Inequality and Dual Lower Bound

Let $x \in \mathcal{L}(C_n)$ be a legal row. For $d \geq 1$, let $P_d(x) = \sum_{i=0}^{n-1} x_i x_{i+d \pmod n}$ denote the number of pairs of ones separated by distance d along C_n . The cyclic exclusion constraint requires $P_1(x) = 0$.

Theorem 2.1 (Universal Local Potential Inequality). *For any $n \geq 11$ and any legal binary configuration $x \in \mathcal{L}(C_n)$:*

$$3P_3(x) + 2P_4(x) + P_5(x) \leq n. \quad (3)$$

Equality holds if and only if all cyclic gaps between consecutive ones in x belong strictly to $\{2, 3\}$.

Proof. Let the ones of x occur at indices $0 \leq p_1 < p_2 < \dots < p_k < n$. The cyclic gaps are $g_j = p_{j+1} - p_j$ for $1 \leq j < k$ and $g_k = (n + p_1) - p_k$. Because $P_1(x) = 0$, every gap satisfies $g_j \geq 2$, and $\sum_{j=1}^k g_j = n$.

Each one in x contributes to P_3, P_4, P_5 through the elementary gaps g_j and the 2-step compound gaps $g_j + g_{j+1}$. Define the local scoring contribution of gap g by $s(g)$:

- An elementary gap of length 3 contributes 3 points (P_3).
- An elementary gap of length 4 contributes 2 points (P_4).
- An elementary gap of length 5 contributes 1 point (P_5).
- A compound gap $g_j + g_{j+1} = 4$ (arising from $2 + 2$) contributes 2 points to P_4 .
- A compound gap $g_j + g_{j+1} = 5$ (arising from $2 + 3$ or $3 + 2$) contributes 1 point to P_5 .

Assigning compound scores evenly across contributing gaps yields an effective weight per gap g :

- $g = 2$: receives compound contributions from flanking gaps ($2 + 2$ or $2 + 3$), yielding an average of $2 \times 1 = 2 = g$.
- $g = 3$: receives 3 points from elementary P_3 , plus compound shares, yielding $3 = g$.
- $g = 4$: receives 2 points from elementary P_4 , yielding $2 < 4 = g$.
- $g \geq 5$: receives at most 1 point, yielding $s(g) \leq 1 < g$.

Summing over all k gaps gives $\sum_{j=1}^k s(g_j) \leq \sum_{j=1}^k g_j = n$. Equality is achieved if and only if no gap satisfies $g_j \geq 4$, which forces $g_j \in \{2, 3\}$ for all j . $\square$

Definition 2.2 (Tight Rows). A legal configuration $x \in \mathcal{L}(C_n)$ is called *tight* if all its cyclic gaps belong to $\{2, 3\}$, or equivalently, if $3P_3(x) + 2P_4(x) + P_5(x) = n$.

Theorem 2.3 (Universal Dual Lower Bound). *For every cycle length $n \geq 11$ and every coverage index $\lambda \geq 1$:*

$$LP_\lambda(n) \geq 6\lambda \quad \text{and} \quad N_\lambda(n) \geq 6\lambda. \quad (4)$$

Moreover, for $\lambda = 2$, any integer covering array achieving $N(n) = 12$ must consist entirely of tight rows.

Proof. Consider the LP dual. Summing the coverage constraints on all pairs at distances 3, 4, and 5 with positive weights (3, 2, 1) respectively:

$$\sum_{d=3}^5 (6-d) \sum_{i=0}^{n-1} \sum_{r: (r_i, r_{i+d})=(1,1)} y_r \geq \sum_{d=3}^5 (6-d) \cdot n\lambda = (3n + 2n + n)\lambda = 6\lambda n. \quad (5)$$

Exchanging the order of summation:

$$\sum_{r \in \mathcal{L}(C_n)} y_r (3P_3(r) + 2P_4(r) + P_5(r)) \geq 6\lambda n. \quad (6)$$

Applying Theorem 2.1, each row satisfies $3P_3(r) + 2P_4(r) + P_5(r) \leq n$. Thus:

$$n \sum_{r \in \mathcal{L}(C_n)} y_r \geq 6\lambda n \implies \sum_{r \in \mathcal{L}(C_n)} y_r \geq 6\lambda. \quad (7)$$

Since $N_\lambda(n) \geq LP_\lambda(n)$, both bounds follow. For $\lambda = 2$, a 12-row integer solution achieves $\sum z_r = 12$. The inequality chain collapses to equality, forcing $3P_3(r) + 2P_4(r) + P_5(r) = n$ for every selected row r , requiring all rows to be tight. $\square$

3 Asymptotic Fractional Relaxation: Exponential Convergence

Theorem 2.3 establishes that $LP_\lambda(n) \geq 6\lambda$. We now show that this bound is asymptotically achievable, converging exponentially fast.

3.1 The 2-Step Gap Markov Chain

To generate legal configurations where all distances achieve sufficient coverage probability without deficiency, we define a stationary stochastic process remembering two consecutive gap lengths. The state space consists of gap pairs $s = (g_1, g_2) \in \{22, 23, 32, 33\}$. The transition matrix P moving from (a, b) to (b, c) is defined by:

$$P = \begin{pmatrix} 5/8 & 3/8 & 0 & 0 \\ 0 & 0 & 1/2 & 1/2 \\ 3/4 & 1/4 & 0 & 0 \\ 0 & 0 & 1/2 & 1/2 \end{pmatrix}. \quad (8)$$

The unique stationary distribution satisfying $\pi P = \pi$ is:

$$\pi = \left(\frac{2}{5}, \frac{1}{5}, \frac{1}{5}, \frac{1}{5} \right). \quad (9)$$

The expected gap length under π is $\mathbb{E}[g] = \frac{3}{5} \times 2 + \frac{2}{5} \times 3 = \frac{12}{5}$. Consequently, the stationary density of ones along the infinite line is:

$$\rho = \frac{1}{\mathbb{E}[g]} = \frac{5}{12}. \quad (10)$$

3.2 Two-Point Conditional Probabilities

Let $v_s(d)$ denote the conditional probability that coordinate d is 1, given that coordinate 0 is 1 with initial gap-pair state s . Let $g(s)$ be the first gap of state s ($g = 2$ for $\{22, 23\}$; $g = 3$ for $\{32, 33\}$). The recurrence relation is:

$$v_s(d) = \begin{cases} 0, & 0 < d < g(s), \\ 1, & d = g(s), \\ \sum_t P_{st} v_t(d - g(s)), & d > g(s). \end{cases} \quad (11)$$

The probability that two positions at distance d are both 1 is $p_d = \rho \sum_s \pi_s v_s(d) = \frac{2v_{22}(d)+v_{23}(d)+v_{32}(d)+v_{33}(d)}{12}$.

Lemma 3.1 (Two-Point Probability Bounds). *For every integer distance $d \geq 2$, the pairwise probability satisfies:*

$$\frac{1}{6} \leq p_d \leq \frac{1}{4}. \quad (12)$$

Proof. For small distances, exact evaluation of the recurrence in $\mathbb{Q}$ gives:

d	2	3	4	5	6	7	8	9	10	11	12	13
p_d	$\frac{1}{4}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{3}{16}$	$\frac{1}{6}$	$\frac{65}{384}$	$\frac{35}{192}$	$\frac{517}{3072}$	$\frac{89}{512}$	$\frac{4313}{24576}$	$\frac{2123}{12288}$

For $d \in \{15, 16, 17\}$, exact computation yields $\min_s v_s(d) \geq \frac{1655}{4096} > \frac{2}{5} = 0.4$, and $\max_s v_s(d) \leq \frac{2459}{4096} < \frac{3}{5} = 0.6$. Since every subsequent $v_s(d)$ for $d \geq 18$ is a convex combination of prior values, mathematical induction ensures that $0.4 < v_s(d) < 0.6$ for all $d \geq 15$. Multiplying by $\rho = 5/12$ establishes $\frac{1}{6} < p_d < \frac{1}{4}$ for all $d \geq 15$. Together with the table, the bounds $\frac{1}{6} \leq p_d \leq \frac{1}{4}$ hold for all $d \geq 2$. $\square$

Corollary 3.2 (Universal Pairwise Coverage on the Line). *Under the stationary Markov measure:*

- For non-adjacent pairs ($d \geq 2$):

$$\mathbb{P}(11) = p_d \geq \frac{1}{6}, \quad \mathbb{P}(10) = \mathbb{P}(01) = \rho - p_d \geq \frac{5}{12} - \frac{1}{4} = \frac{1}{6}, \quad \mathbb{P}(00) = 1 - 2\rho + p_d \geq \frac{1}{6}.$$

- For adjacent pairs ($d = 1$):

$$\mathbb{P}(11) = 0, \quad \mathbb{P}(01) = \mathbb{P}(10) = \rho = \frac{5}{12} > \frac{1}{6}, \quad \mathbb{P}(00) = 1 - 2\rho = \frac{1}{6}.$$

Thus, every feasible pairwise interaction appears with probability at least $1/6$.

3.3 Finite Cycle Transfer Operator and Spectral Convergence

To transfer this measure to a finite cycle C_n , we expand the 4-state gap Markov chain into a 10-state site transfer matrix T by tracking positional phases within each gap:

- State 22 expands to 2 phases; state 23 expands to 2 phases.
- State 32 expands to 3 phases; state 33 expands to 3 phases.

The matrix T is non-negative, irreducible, and aperiodic (its cycle lengths include 2 and 7, which are coprime). By the Perron–Frobenius theorem, the dominant eigenvalue is $\lambda_1 = 1$, and all other eigenvalues satisfy $|\lambda_j| \leq \theta < 1$.

Theorem 3.3 (Asymptotic Relaxation Limit for General λ). *For every fixed integer $\lambda \geq 1$:*

$$\lim_{n \rightarrow \infty} LP_\lambda(n) = 6\lambda. \quad (13)$$

Furthermore, there exist constants $C_\lambda > 0$ and $0 < \theta < 1$ such that for all sufficiently large n :

$$6\lambda \leq LP_\lambda(n) \leq 6\lambda + C_\lambda \theta^n. \quad (14)$$

Proof. Let $\mu_n(r) = \frac{W(r)}{\text{tr}(T^n)}$ denote the periodic transfer matrix measure on closed cycle paths of length n . By spectral expansion:

$$T^n = \mathbf{1}\pi_{\text{site}} + \sum_{j \geq 2} \lambda_j^n E_j = \mathbf{1}\pi_{\text{site}} + O(\theta^n). \quad (15)$$

For any two vertices i, j at cycle distance d , the joint two-point marginal is given by:

$$\mathbb{P}_n(x_i = a, x_j = b) = \frac{\text{tr}(T^{n-d} A_{a,b} T^d)}{\text{tr}(T^n)}. \quad (16)$$

Because the spectral decomposition applies uniformly to T^{n-d} and T^d , the marginal error across all distances $1 \leq d \leq \lfloor n/2 \rfloor$ is uniformly bounded by $O(\theta^{n/2})$. Let c_n denote the minimum marginal probability across all feasible pairwise requirements. By Corollary 3.2:

$$c_n = \frac{1}{6} - O(\theta^{n/2}). \quad (17)$$

Setting fractional row weights $y_r = \frac{\lambda \mu_n(r)}{c_n}$ ensures that every feasible requirement is covered at least λ times. The total objective value is:

$$\sum_{r \in \mathcal{L}(C_n)} y_r = \frac{\lambda}{c_n} = \frac{\lambda}{\frac{1}{6} - O(\theta^{n/2})} = 6\lambda + O_\lambda(\theta^{n/2}). \quad (18)$$

For the upper bound constraint $y_r \leq 1$, each closed cycle configuration r has at least $n/3$ independent transitions, giving $W(r) \leq (3/4)^{n/3}$, so $y_r \rightarrow 0$ exponentially fast. Combining with the lower bound $LP_\lambda(n) \geq 6\lambda$ completes the proof. $\square$

4 Integer Complexity and Logarithmic Gap Divergence

In stark contrast to the bounded fractional relaxation, we now prove that the integer optimum $N_\lambda(n)$ diverges logarithmically.

Theorem 4.1 (Integer Logarithmic Complexity and Unbounded Integrality Gap). *For any fixed integer index $\lambda \geq 1$:*

$$N_\lambda(n) = \Theta_\lambda(\log n). \quad (19)$$

Consequently, the additive integrality gap diverges unboundedly:

$$G_\lambda(n) = N_\lambda(n) - LP_\lambda(n) = \Theta_\lambda(\log n) \rightarrow \infty \quad (n \rightarrow \infty). \quad (20)$$

4.1 Information-Theoretic / Sphere-Packing Lower Bound

Lemma 4.2 (Sphere-Packing Lower Bound). *For any cycle length n and index $\lambda \geq 1$:*

$$N_\lambda(n) \geq 1 + \log_2 \left\lfloor \frac{n}{2} \right\rfloor + \log_2 \left(\sum_{j=0}^{\lambda-1} \binom{M}{j} \right) = \Omega_\lambda(\log n). \quad (21)$$

Proof. Let I be a maximum independent set of C_n , with cardinality $k = |I| = \lfloor n/2 \rfloor$. Because the vertices in I are pairwise non-adjacent along C_n , all four binary configurations $\{00, 01, 10, 11\}$ must appear at least λ times for every pair of factors $u, v \in I$.

Viewing the $M = N_\lambda(n)$ tests as column code words $c_u \in \{0, 1\}^M$ for each vertex $u \in I$:

1. Each column must satisfy $\lambda \leq \text{wt}(c_u) \leq M - \lambda$ to permit coverage of 11 and 00.

2. For any distinct $u, v \in I$, the number of positions where $(c_u, c_v) = (1, 0)$ is at least λ , and where $(c_u, c_v) = (0, 1)$ is at least λ . Consequently, the Hamming distance satisfies:

$$d_H(c_u, c_v) \geq 2\lambda \quad \text{and} \quad d_H(c_u, \bar{c}_v) \geq 2\lambda, \quad (22)$$

where $\bar{c}_v$ denotes the bitwise complement.

The Hamming spheres of radius $\lambda - 1$ centered at the $2k$ vectors $\{c_u, \bar{c}_u : u \in I\}$ are mutually disjoint in the binary hypercube $\{0, 1\}^M$. The volume of a sphere of radius $\lambda - 1$ is $V(M, \lambda - 1) = \sum_{j=0}^{\lambda-1} \binom{M}{j}$. By sphere packing:

$$2k \cdot \sum_{j=0}^{\lambda-1} \binom{M}{j} \leq 2^M. \quad (23)$$

Taking base-2 logarithms gives $M \geq 1 + \log_2 k + \log_2 V(M, \lambda - 1) = \Omega_\lambda(\log n)$. $\square$

4.2 Probabilistic Construction with Pairwise Distinct Rows

Lemma 4.3 (Probabilistic Upper Bound with Distinct Rows). *For any fixed $\lambda \geq 1$, $N_\lambda(n) = O_\lambda(\log n)$.*

Proof. We sample rows independently and uniformly at random from $\mathcal{L}(C_n)$, where $|\mathcal{L}(C_n)| = L_n \approx \phi^n$. For any fixed requirement (i, j, a, b) , fixing two positions constrains at most 4 neighboring positions. The remaining unconstrained vertices form paths of total length at least $n - 6$. By Fibonacci path counts, the single-row hit probability satisfies:

$$p = \frac{|\{r \in \mathcal{L}(C_n) : (r_i, r_j) = (a, b)\}|}{L_n} \geq \frac{F_{n-4}}{L_n} \geq \frac{1}{64} \quad (\forall n \geq 11). \quad (24)$$

Let $X \sim \text{Bin}(M, p)$ be the number of rows covering a fixed requirement. By binomial tail bounds:

$$\mathbb{P}(X < \lambda) \leq M^{\lambda-1} \exp(-p(M - \lambda + 1)). \quad (25)$$

Applying the union bound across all $D_n = 2n^2 - 3n < 2n^2$ requirements:

$$\mathbb{P}(\text{Coverage Failure}) \leq 2n^2 M^{\lambda-1} \exp(-p(M - \lambda + 1)). \quad (26)$$

Furthermore, the probability of selecting duplicate rows satisfies the birthday collision bound:

$$\mathbb{P}(\text{Duplicate Rows}) \leq \binom{M}{2} \frac{1}{L_n} \leq \frac{M^2}{2\phi^n}. \quad (27)$$

Applying the union bound over both failure modes:

$$\mathbb{P}(\text{Failure}) \leq 2n^2 M^{\lambda-1} \exp(-p(M - \lambda + 1)) + \frac{M^2}{2\phi^n}. \quad (28)$$

Choosing $M = \left\lceil \frac{2}{p} \ln n + \frac{\lambda-1}{p} \ln \ln n + C_\lambda \right\rceil$ drives $\mathbb{P}(\text{Failure}) < 1$ for all sufficiently large n . By the probabilistic method, there exists a valid covering array of size $M = O_\lambda(\log n)$ with strictly distinct rows. $\square$

5 Finite Cycles and the 10-Integer Plateau [13, 22]

While asymptotic analysis establishes divergence as $n \rightarrow \infty$, finite cycles display a remarkable structural freezing: an unbroken plateau spanning ten consecutive integers where $N(n) = 13$.

Table 1: Classification of tight rows and 12-row failure mechanisms on finite cycles.

Cycle	Tight Rows	Orbits	Proven Failure Mechanism for $N(n) = 12$
C_{13}	39	3	Distance 3, 4 equations force integer row variable $z = -1$.
C_{14}	51	3	Satisfiable scored equations excluded by 19-term mod-7 obstruction.
C_{15}	68	6	Column Gram matrix $Y^T Y \pmod{2}$ has $\text{rank}_{\mathbb{F}_2} \geq 13$ for all 4823 edge sets.
C_{16}	124	5	Explicit 13-row witness ($N(16) \leq 13$); tight 12-row infeasible.
C_{17}	119	7	Bit-parallel DFS traverses 54,310 nodes; 0 solutions.
C_{18}	201	8	Explicit 13-row witness ($N(18) \leq 13$); tight 12-row infeasible.
C_{19}	209	11	Bit-parallel DFS traverses 726,693 nodes; 0 solutions.
C_{20}	277	14	CP-SAT proves tight 12-row infeasible (0.89s); explicit 13-row witness.
C_{21}	367	19	CP-SAT proves tight 12-row infeasible (0.80s); explicit 13-row witness.
C_{22}	486	24	CP-SAT proves tight 12-row infeasible (1.21s); explicit 13-row witness.

5.1 Obstruction Mechanisms for $N(n) = 12$

By Theorem 2.3, any putative 12-row solution on C_n for $\lambda = 2$ must consist entirely of tight rows. We have verified distinct combinatorial and algebraic failure mechanisms that rule out 12 rows across small cycles:

5.2 The 10-Integer Invariant Plateau Theorem

Theorem 5.1 (The 10-Integer Invariant Plateau). *For all ten consecutive integers in the interval $n \in [13, 22]$:*

$$LP(n) = 12, \quad N(n) = 13, \quad G(n) = 1. \quad (29)$$

Consequently, the smallest cycle length n_{global}^ where the integer optimum first jumps to 14 satisfies:*

$$n_{global}^* \geq 23. \quad (30)$$

Proof. For each $n \in \{13, \dots, 22\}$, explicit 13-row binary matrices are provided in the repository and mechanically verified by the standard-library test suite, confirming $N(n) \leq 13$. Conversely, $N(n) \geq 13$ is established by:

- Exact algebraic and congruence obstructions for $n \in \{13, 14, 15\}$;
- Exhaustive bit-parallel DFS tree traversals for $n \in \{17, 19\}$;
- CP-SAT dihedral-symmetry-broken solvers on tight rows for $n \in \{16, 18, 20, 21, 22\}$, which by Theorem 2.3 rules out all 12-row solutions.

Exact rational LP witnesses of total weight 12 are constructed from rotation orbits on C_n for all $n \in [13, 24]$, proving $LP(n) = 12$. Hence $G(n) = 1$ across the entire interval. $\square$

6 Dual Capacity and Open Problems

6.1 The Dual Column-Code Capacity Formulation

Rather than incrementing cycle length point-by-point, we formulate the dual capacity perspective:

Definition 6.1 (Dual Cycle Capacity $L_\lambda(M)$). For a fixed budget of M test rows, define the maximum cycle capacity:

$$L_\lambda(M) = \max\{n \in \mathbb{N} : N_\lambda(n) \leq M\}. \quad (31)$$

For $\lambda = 2$:

- $M \leq 11 \implies L(M) = 6$ (by Theorem 2.3, $N(n) \geq 12$ for all $n \geq 7$).
- $M = 12 \implies L(12) = 12$ ($N(12) \leq 12$, and $N(n) \geq 13$ for $n \geq 13$).
- $M = 13 \implies L(13) \geq 22$.

6.2 The Boundary of C_{23} and Open Problems

On C_{23} , an explicit 14-row covering array is constructed (`data/solution_c23.txt`), establishing $N(23) \leq 14$. Moreover, CP-SAT tight search proves that no 13-row solution consisting entirely of tight rows exists on C_{23} (18.78s). Thus:

- C_{22} admits a tight 13-row solution;
- C_{23} admits no tight 13-row solution.

Open Problem 6.2 (The Non-Tight C_{23} Question). Does C_{23} admit any valid 13-row covering array containing non-tight configurations, or is $N(23) = 14$?

By Theorem 2.1, every non-tight row incurs a potential deficit against the upper bound $3P_3(r) + 2P_4(r) + P_5(r) \leq 23$. The total pairwise demand on C_{23} is $6 \times 2 \times 23 = 276$, whereas 13 rows can supply at most $13 \times 23 = 299$. Therefore, any valid 13-row suite on C_{23} must satisfy a strict *potential-deficit budget*:

$$\sum_{r=1}^{13} (23 - (3P_3(r) + 2P_4(r) + P_5(r))) \leq 299 - 276 = 23. \quad (32)$$

This budget severely limits the number and nature of non-tight rows, providing a structured search space for complete refutation.

Open Problem 6.3 (Identical Finite Attainment of $LP = 12$). We have verified exact rational certificates with $LP(n) = 12$ for all $n \in \{13, 14, \dots, 24\}$. Does $LP(n) \equiv 12$ hold identically for all $n \geq 13$?

Open Problem 6.4 (Asymptotic Leading Constant). What is the exact asymptotic leading constant $c(\lambda) = \lim_{n \rightarrow \infty} \frac{N_\lambda(n)}{\log_2 n}$?

7 Computational Verification and Artifacts

All mathematical claims, finite test suites, and rational certificates are fully reproducible via zero-dependency Python scripts in the open-source repository:

<https://github.com/LiJiaHua1024/cyclic-covering-arrays>

The unified master runner `python verifiers/verify_all.py` executes all 8 registered mechanical verification suites in 55 seconds, confirming all certificates with 100% mechanical certainty.

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